FORECAST INTELLIGENCE AND RCA AGENT

Interview Guide and Reusable Implementation Playbook

Databricks Genie • Delta Tables • Governed Semantic Views • Deterministic Metrics

Purpose: A concise reference for explaining the implemented workflow, the reasoning behind its design, the problems solved, and the implementation pattern to reuse on future analytics projects.

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How to use this guide

- Memorize the 60-second explanation and the end-to-end query flow first.
- Use the architecture, table grains and metric definitions when the interviewer asks technical follow-ups.
- Use the challenge-and-correction section to explain engineering decisions and lessons learned.
- Use the final checklist as a template for future Genie or natural-language analytics projects.

Accuracy boundary: Describe capabilities, architecture and corrections that were implemented. Do not invent user counts, measured adoption, source-system names or business outcomes that cannot be evidenced. If asked about adoption, distinguish a designed business-analyst workflow, a pilot, and production use.

1. The 60-second explanation

"I built a governed Forecast Intelligence and RCA workflow in Databricks Genie for business analysts. Analysts can ask natural-language questions about SKU and customer forecasts, forecast accuracy, approved SKU budgets, pricing, promotions, inventory and forecast deviations. The data is stored in separate Delta fact and dimension tables, while Genie is exposed only to certified semantic views. Critical joins, forecast-vintage selection, aggregation grains and KPI formulas are deterministic in those views; Genie handles intent interpretation, filtering, SQL generation and presentation. I added certification checks for duplicate grains, join cardinality, fiscal periods and effective-dated prices, plus numerical benchmark tests. This gives analysts a conversational interface without allowing the model to invent important business calculations."

Key message to remember

Core design principle: Genie may interpret the question dynamically, but joins, grains, versions and material business calculations are governed deterministically.

2. Business problem and supported analysis

Business need	Implemented capability
Forecast performance	Actual, forecast, error, WMAPE, accuracy and bias by period and entity.
Customer forecast	Customer-SKU performance with customer and SKU entity resolution.
Plan versus actual	Monthly comparison of actual, forecast and approved SKU budget.
Revenue analysis	Gross and net revenue analysis for supported product/store data.
Inventory evidence	As-of balances and qualified inventory exposure; balances are not summed across dates.
Commercial evidence	Effective-dated SKU price and approved promotion records.
RCA support	Evidence-based investigation of associations with inventory, promotion, price and other populated signals.
Follow-up questions	Compatible entity and period context retained within the same conversation.

3. Architecture

Layer	Responsibility	Why it exists
Delta source layer	Separate dimensions and facts at documented grains.	Preserves ownership, refreshability and auditability.
Certification layer	Checks schema, keys, duplicates, references, intervals and publication readiness.	Prevents invalid data from reaching semantic views.
Semantic view layer	Applies approved joins, vintage rules, aggregations and metrics.	Removes critical join decisions from the LLM.
Genie Agent	Interprets natural language, selects a certified view, generates SQL and explains results.	Provides conversational self-service analysis.
Validation layer	Benchmark questions and exact numerical assertions.	Detects regressions and incorrect business semantics.

Architecture flow to draw in an interview

Source data → Delta facts/dimensions → Certification → Semantic views → Genie SQL → Business answer

4. Most important: query-to-answer flow

1. A business analyst asks a natural-language question in the Genie conversation.
2. Genie identifies the intent: SKU forecast, customer forecast, budget, price, promotion, revenue, inventory or RCA.
3. Genie resolves names and IDs through the governed entity lookup when necessary.
4. Conversation rules determine whether the question is a follow-up or an explicit entity/period switch.
5. Genie selects the narrowest certified semantic view that already represents the required business grain.
6. Genie generates SQL with filters, grouping, sorting and limits against that view.
7. Databricks SQL applies access permissions and executes the query.
8. The semantic view supplies deterministic joins, latest-vintage selection, fiscal mapping and certified metrics.
9. The result is returned to Genie as structured rows.
10. Genie formats the result using business labels, units, currency, percentages and period boundaries.
11. For RCA, Genie describes supported signals as associated evidence and avoids causal claims.
12. The analyst can ask a follow-up; compatible context is retained only within that conversation.

What is deterministic and what is dynamic?

Deterministic	Dynamic within guardrails
Table grain and uniqueness	Natural-language intent interpretation
Fact-to-dimension joins	Choice of certified view
Latest forecast vintage per target	Filters requested by the user
Fiscal-period mapping	Sorting, top-N and presentation
Metric formulas and denominators	Compatible follow-up context
Budget/version and price-validity rules	Plain-language explanation
Certification and blocking conditions	SQL shape within the semantic contract

5. Tables and views to know

Physical tables

Object	One row represents
dim_sku	One globally unique SKU with descriptive attributes.
dim_customer	One globally unique customer with segment and geography.
dim_fiscal_period	One country and fiscal period with start/end dates.
fact_sku_forecast	Region, country, SKU, target date and forecast vintage.
fact_customer_forecast	Region, country, customer, SKU, target date and forecast vintage.
fact_sku_budget	Region, country, SKU, fiscal period and budget version.
fact_sku_price	Country, SKU, price type and effective-from date.
fact_sku_promotion	Region, country, SKU and promotion ID.

Genie-facing semantic views

View	Purpose
vw_genie_entity_lookup	Resolve SKU/customer names and entity types.
vw_genie_sku_forecast	Detailed SKU forecast and actual rows.
vw_genie_customer_forecast	Detailed customer-SKU forecast and actual rows.
vw_genie_sku_budget	Approved SKU quantity and revenue budgets.
vw_genie_sku_price	Effective-dated SKU list prices.
vw_genie_sku_promotion	Approved SKU promotion events.
vw_genie_sku_monthly_performance	Certified monthly SKU forecast metrics.
vw_genie_customer_sku_monthly_performance	Certified monthly customer-SKU metrics.
vw_genie_sku_monthly_plan_vs_actual	Certified monthly actual, forecast and budget comparison.
vw_genie_forecast_observations	Original product/store forecast, revenue, inventory and RCA evidence.

Why facts are not exposed to Genie: If facts and semantic views are both exposed, Genie can select an unsafe path, recreate joins, multiply rows or combine forecast versions. The views are the governed analytical contract.

6. Join and aggregation logic

Join rules

- Fact-to-dimension relationships are many-to-one and use stable entity keys.
- Fact row counts must equal enriched base-view row counts; a mismatch identifies an orphan or many-to-many join.
- Raw forecast, budget, price and promotion facts are not dynamically joined by Genie.
- Cross-domain questions use a pre-certified view at a common grain.
- SELECT DISTINCT is not used to hide duplicate results; duplicate grains block publication.

Aggregation rules

Measure type	Examples	Rule
Additive	Quantity, revenue	Sum only across compatible units/currency and approved dimensions.
Semi-additive	Inventory balance	Sum across entities at one as-of point;

		never across dates.
Non-additive	Price, percentages	Select effective value or recompute from base totals.
Versioned	Forecast, budget	Select the approved version/vintage before aggregation.
Effective-dated	Price, promotion	Match the requested date to the valid interval.

7. Metric definitions

Metric	Definition / rule
Forecast error	Forecast − actual.
Absolute error	ABS(forecast − actual).
WMAPE	100 × SUM(absolute error) ÷ SUM(ABS(actual)).
Forecast accuracy	100 − WMAPE; do not silently clamp the result.
Bias %	100 × SUM(forecast − actual) ÷ SUM(ABS(actual)).
Forecast vs actual	Forecast − actual; percentage denominator is actual.
Budget vs actual	Budget − actual; percentage denominator is actual.
Forecast vs budget	Forecast − budget; percentage denominator is budget.
Zero denominator	Return unavailable/null rather than inventing zero.

Common mistake: Never average row-level WMAPE or accuracy. Aggregate numerator and denominator first, then calculate the percentage.

8. Genie instructions and examples

Essential instruction categories

- Source routing: which certified view supports each type of question.
- Semantic safety: no raw facts, unsafe joins, DISTINCT-based repair or mixed units/currencies.
- Context: retain compatible scope; explicit new entity or period replaces old context.
- Metric definitions: approved formulas, signs and denominators.
- RCA language: association is evidence; it is not automatically causation.
- Presentation: business labels, compact tables, units, currency, percentages and evidence limitations.

Why verified question-SQL examples matter

- They teach Genie which semantic view to choose for a recurring business pattern.
- They demonstrate the expected fields, filters, grouping and ordering.
- They reduce the risk of re-creating calculations from detailed rows.
- They are guidance; Genie can still generate new SQL for variations in entity, period, ranking and comparison.

Do not over-prompt: Keep the main instruction concise. Put stable business logic in views and use verified examples for query patterns. Prompts should describe behavior, not replace the semantic model.

9. Conversation behavior

User behavior	Expected handling
“How accurate was that forecast?”	Retain compatible entity and period from the current conversation.
Names a different SKU	Switch SKU while preserving compatible period/geography.
Changes from forecast to budget	Retain SKU only if budget is valid for that entity.
Asks for customer budget	Explain that customer budget is unsupported; do not allocate SKU budget.
Starts a new conversation	Do not inherit entity or period from another conversation.
Uses an ambiguous name	Resolve through entity lookup or ask one clarification question.

10. Challenges faced and corrections implemented

Observed challenge	Correction implemented	Lesson
Dynamic joins could create untrusted totals.	Exposed certified semantic views and kept raw facts outside Genie.	Critical joins belong in governed SQL, not prompts.
Weekly forecast and monthly budget used different grains.	Added fiscal-period mapping and a monthly plan-versus-actual view.	Aggregate each source to a common grain before combining.
Multiple forecast vintages could double-count results.	Applied LATEST_PER_TARGET before monthly aggregation.	Version selection must precede aggregation.
Join multiplication could pass unnoticed.	Added fact/view row-count and semantic-grain certification.	Validate cardinality instead of using DISTINCT.
Percentages could be averaged or use inconsistent denominators.	Moved certified metrics and signed variances into semantic views.	Store metric semantics close to data.
Price records could overlap.	Added effective-date overlap validation.	Effective-dated facts require interval certification.
Fiscal periods could overlap or fail to map.	Added explicit fiscal-period dimension and overlap checks.	Calendar logic must be data, not inferred text.
Customer requests could incorrectly receive SKU budget.	Added entity routing and explicit unsupported behavior.	Unsupported analysis should fail clearly.
Follow-ups could lose or incorrectly retain context.	Added compatible-context retention and explicit entity switching.	Conversation state needs controlled replacement rules.
RCA wording could imply causality.	Restricted conclusions to supported associations and evidence coverage.	Analytical correlation is not causal proof.
Results were too technical or poorly formatted.	Added business labels, Markdown tables, units and signed deltas.	Presentation is part of analytical correctness.
Large instruction text reduced clarity.	Condensed instructions and moved query patterns into examples/views.	Use prompts for policy and semantic models for logic.

11. Validation approach

- Data validation: required columns, non-empty keys, compatible units/currency and referential integrity.
- Grain validation: no duplicates at each documented fact or semantic-view grain.
- Cardinality validation: enriched view row count equals source fact row count.
- Version validation: one selected forecast vintage per target before aggregation.
- Interval validation: fiscal periods and effective-dated prices do not overlap.
- Numerical validation: benchmark questions assert exact actual, forecast, budget, WMAPE, accuracy, bias and deltas.
- Behavioral validation: follow-up context, entity switching, unsupported requests and RCA qualification.

Example benchmark to remember

Scope	Expected result
SKU-1001, FY2026-P08	Actual 635; forecast 565; approved budget 480.
Forecast vs actual	−70; approximately −11.0% of actual.
Budget vs actual	−155; approximately −24.4% of actual.
Forecast vs budget	+85; approximately +17.7% of budget.
Policy/version	LATEST_PER_TARGET; budget version FY2026-V1.

12. Interview questions and concise answers

Why Genie rather than a custom LLM application?

Genie provides natural-language-to-SQL, conversational context and Databricks governance close to the data. I still constrained it with semantic views because natural-language generation alone is not a business-control layer.

Is this agentic AI?

It has agent-like interpretation, tool selection, SQL generation and multi-turn behavior. I describe it precisely as a governed Genie Agent workflow rather than claiming a fully autonomous multi-agent system.

How did you prevent hallucinated numbers?

The model does not create source facts. Numbers come from certified views, deterministic formulas and SQL execution. Unsupported evidence returns an explicit unavailable or blocked status.

How did you prevent wrong joins?

Genie sees certified views, not raw facts. Cross-domain joins are implemented once at an approved common grain and then tested for cardinality and exact results.

How does it handle follow-ups?

It retains compatible entity and period context inside one conversation. Explicit new entities or periods replace the old scope, and ambiguous requests trigger clarification.

How is RCA performed?

The workflow retrieves qualified evidence such as forecast error, inventory, price and promotion. It reports supported associations and coverage limitations; it does not claim causality without causal evidence.

What happens if data is missing?

Missing data remains null or unavailable. Partial coverage is disclosed. Certification failures block the view rather than silently returning misleading totals.

How do you test an LLM-generated SQL system?

I combine deterministic data and view tests with exact benchmark questions, generated-SQL inspection, negative tests and conversation tests.

What did you learn?

The main lesson was to move joins, grain and metrics out of prompts and into a governed semantic layer. Prompts guide behavior; they should not carry the entire analytical model.

13. Reusable implementation blueprint

13. Document business questions, entities, measures, periods and required evidence.
14. Define every fact table grain before writing transformations or prompts.
15. Create stable dimension keys and validate many-to-one relationships.
16. Classify measures as additive, semi-additive, non-additive, versioned or effective-dated.
17. Build narrow semantic views for recurring business question families.
18. Aggregate facts independently to a certified common grain before cross-domain joins.
19. Encode versions, calendars, denominators and null behavior in SQL.
20. Add certification gates for schema, keys, duplicates, references, intervals and cardinality.
21. Expose only certified views to the natural-language agent.
22. Write concise routing and safety instructions.
23. Add verified question-SQL examples for high-value patterns.
24. Create exact numerical benchmarks and negative tests.
25. Inspect generated SQL for novel questions and promote repeated patterns into semantic views.

14. Final memory sheet

One-line architecture: Delta facts and dimensions → certification → governed semantic views → Genie-generated SQL → formatted business answer.

One-line governance rule: Genie chooses filters and presentation; certified SQL controls joins, grain, versions and formulas.

One-line RCA rule: Report evidence and association, never unsupported causation.

One-line testing rule: Validate both the numerical result and the SQL path used to obtain it.

Terms to remember

Term	Meaning
Grain	The exact business meaning of one row.
Cardinality	The number relationship between joined datasets; incorrect cardinality multiplies or loses rows.
Semantic view	A governed analytical contract with approved joins, fields and calculations.
Forecast vintage	The version of a forecast defined by its generation date.
WMAPE	Total absolute error divided by total absolute actual demand.
Semi-additive	A measure that can be summed across some dimensions but not time.
Effective-dated	A record valid only between defined start and end dates.
Certification	Automated proof that a dataset meets required schema and business rules.

Safe wording for the interview

"I implemented a business-analyst-facing Genie workflow and governed its analytical behavior through certified Delta views, deterministic metrics and regression benchmarks. The important engineering decision was to prevent the language model from inventing critical joins or aggregation logic."

If asked for adoption evidence: Answer with the exact scope you can support: implemented workflow, hosted demonstration, pilot, or production. Do not substitute an assumed user count for evidence.